STA 445 S26 Assignment 5

2026-03-19

Problem 1

Convert the following to date or date/time objects.

a. September 13, 2010.

```
mdy("September 13, 2010")
```

```
## [1] "2010-09-13"
```

b. Sept 13, 2010.

```
mdy("Sept 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

Sept isn't recognized as September.

c. Sep 13, 2010.

```
mdy("Sep 13, 2010")
```

```
## [1] "2010-09-13"
```

d. S 13, 2010. Comment on the month abbreviation needs.

```
mdy("S 13, 2010")
```

```
## Warning: All formats failed to parse. No formats found.
```

```
## [1] NA
```

S isn't recognized as September.

e. 07-Dec-1941.

```r
dmy("07-Dec-1941")
```

```
## [1] "1941-12-07"
```

f. 1-5-1998. Comment on why you might be wrong.

```r
dmy("1-5-1988")
```

```
## [1] "1988-05-01"
```

Whoever uses the "Day-Month-Year" format would read this as January 5th, 1988 but to others it could be read as May 1st, 1988.

g. 21-5-1998. Comment on why you know you are correct.

```r
dmy("21-5-1988")
```

```
## [1] "1988-05-21"
```

this will be may 21st because there isn't 21 months so this will make sense to anyone reading it and they would assume it falls under the "Day-Month-Year" format.

h. 2020-May-5 10:30 am

```r
ymd_hm("2020-May-5 10:30 am")
```

```
## [1] "2020-05-05 10:30:00 UTC"
```

i. 2020-May-5 10:30 am PDT (ex Seattle)

```r
ymd_hm("2020-May-5 10:30 am", tz = "US/Pacific")
```

```
## [1] "2020-05-05 10:30:00 PDT"
```

j. 2020-May-5 10:30 am AST (ex Puerto Rico)

```r
ymd_hm("2020-May-5 10:30 am", tz = "America/Puerto_Rico")
```

```
## [1] "2020-05-05 10:30:00 AST"
```

Problem 2

Using just your date of birth (ex Sep 7, 1998) and today's date calculate the following:

```r
MB <- mdy("01/04/2000")
```

a. Calculate the date of your 64th birthday.

```r
MB + years(64)
```

```
## [1] "2064-01-04"
```

b. Calculate your current age (in years).

```r
life <- MB %--% today()
current_age <- as.period(life, unit = "years")
year(current_age)
```

```
## [1] 26
```

c. Using your result in part (b), calculate the date of your next birthday.

```r
nextBday <- MB + years(year(current_age) + 1)
nextBday
```

```
## [1] "2027-01-04"
```

d. The number of *days* until your next birthday.

```r
until_bday <- today() %--% nextBday
as.numeric(as.duration(until_bday), "days")
```

```
## [1] 291
```

e. The number of *months* and *days* until your next birthday.

```r
as.period(until_bday, unit = "months")
```

```
## [1] "9m 16d 0H 0M 0S"
```

Problem 3

Suppose you have arranged for a phone call to be at 3 pm on May 8, 2015 at Arizona time. However, the recipient will be in Auckland, NZ. What time will it be there?

```r
AZ <- ymd_hm("2015-05-08 15:00", tz = "US/Arizona")

Auckland <- with_tz(AZ, tzone = "Pacific/Auckland")


Auckland
```

```
## [1] "2015-05-09 10:00:00 NZST"
```

it will be 10am in Auckland on May 9th.

Problem 4

It turns out there is some interesting periodicity regarding the number of births on particular days of the year.

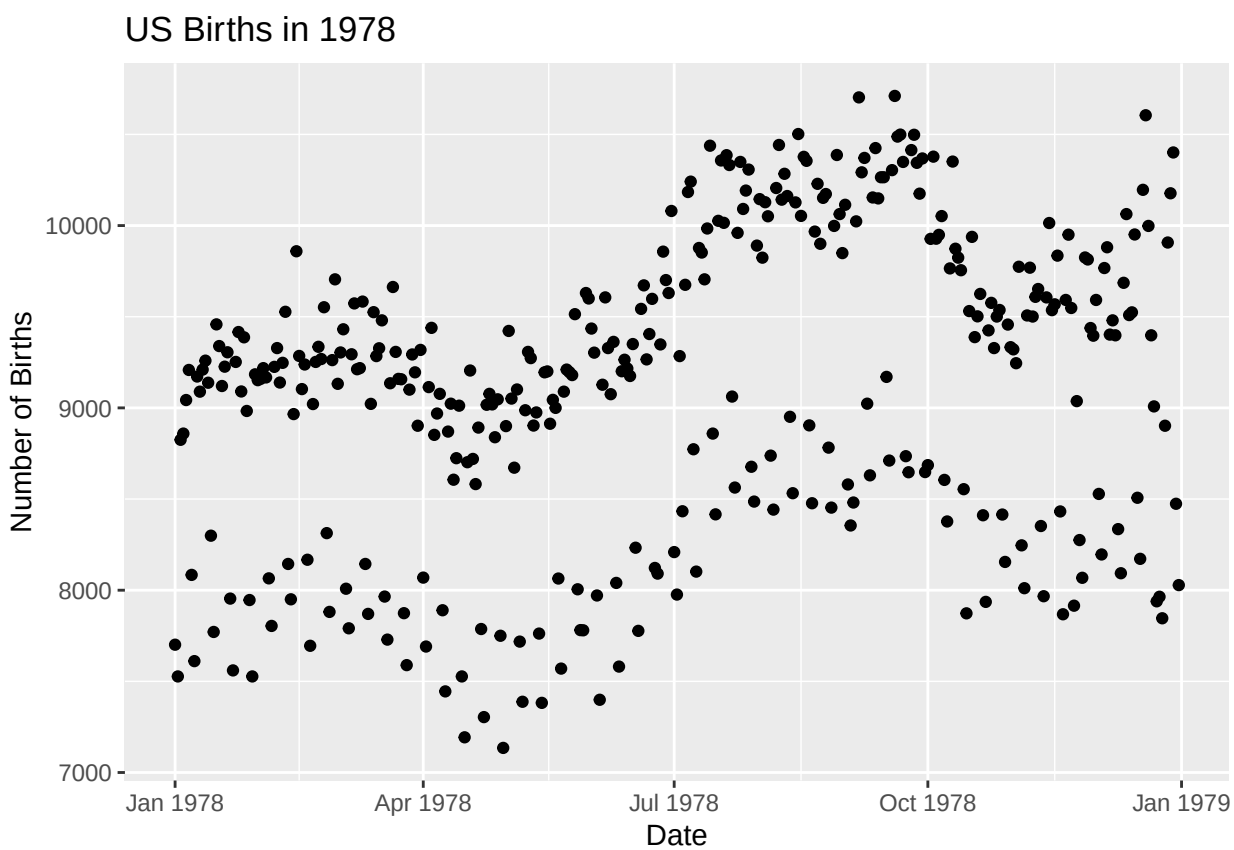
a. Using the `mosaicData` package, load the data set `Births78` which records the number of children born on each day in the United States in 1978. Because this problem is intended to show how to calculate the information using the `date`, remove all the columns *except* `date` and `births`.

```r
library(mosaicData)
library(tidyverse)
library(lubridate)

births_new <- Births78 %>%
  select(date, births)

head(births_new)
```

```
##         date births
## 1 1978-01-01   7701
## 2 1978-01-02   7527
## 3 1978-01-03   8825
## 4 1978-01-04   8859
## 5 1978-01-05   9043
## 6 1978-01-06   9208
```

b. Graph the number of `births` vs the `date` with date on the x-axis. What stands out to you? Why do you think we have this trend?

```r
ggplot(births_new, aes(x = date, y = births)) +
  geom_point() +
  labs(title = "US Births in 1978", x = "Date", y = "Number of Births")
```

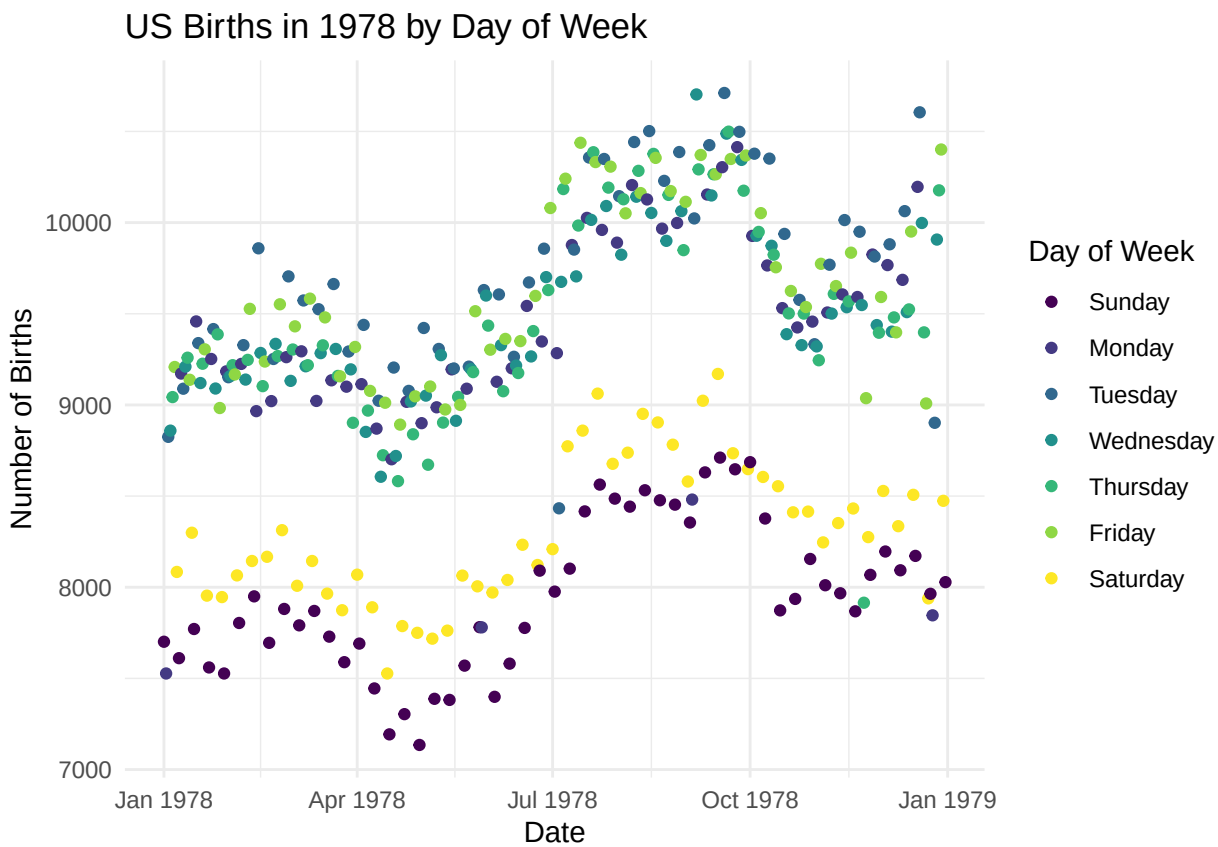


c. To test your assumption, we need to figure out the what day of the week each observation is. Use `dplyr::mutate` to add a new column named `dow` that is the day of the week (Monday, Tuesday, etc). This calculation will involve some function in the `lubridate` package and the `date` column.

```
births_new<- births_new %>%
  mutate(dow = wday(date, label = TRUE, abbr = FALSE))

head(births_new)
```

```
##         date births       dow
## 1 1978-01-01   7701    Sunday
## 2 1978-01-02   7527    Monday
## 3 1978-01-03   8825   Tuesday
## 4 1978-01-04   8859 Wednesday
## 5 1978-01-05   9043  Thursday
## 6 1978-01-06   9208    Friday
```

d. Plot the data with the point color being determined by the day of the week variable.

```
ggplot(births_new, aes(x = date, y = births, color = dow)) +
  geom_point() +
  labs(title = "US Births in 1978 by Day of Week",
       x = "Date",
```

```r
      y = "Number of Births",
      color = "Day of Week") +
theme_minimal()
```